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Chemistry 12

Lab 2B: Effect of Concentration on Equilibrium Systems

$$Fe^{3+} + SCN^- \rightleftharpoons FeSCN^{2+}$$

Fe^{3+}	SCN^-	$FeSCN^{2+}$
(pale yellow)	(colourless)	(blood red)

1 Purpose

To explore the effect of concentration change in the reaction $Fe^{3+} + SCN^- \rightleftharpoons FeSCN^{2+}$.

2 Procedure

See ***Essential Experiments for Chemistry - Lab 12A, Part II (p.165)*** for the procedure. Note that the initial preparation beaker was split across two groups of students.

3 Data/Observations

The initial solution of liquid appeared a light orange. After the addition of each of the reagents, the colour of the solutions in test tubes B, C, and E changed.

Reagent	Stress	Colour observation	Shift
Control (A)		yellow-orange	
KSCN (B)		orange-red	
$FeCl_3$ (C)		orange-red	
KCl (D)		yellow-orange	
NaOH (E)		pale yellow	

Table 1: Reagent additions and colour observations

4 Analysis

5 Conclusion